Access Control in Snowflake

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Agenda

- What is Access Control
- Key Concepts of Access Control
- Objects Hierarchy
- Roles in Snowflake
- System Defined Roles
- Role Hierarchy
- Custom Roles

Access Control in Snowflake

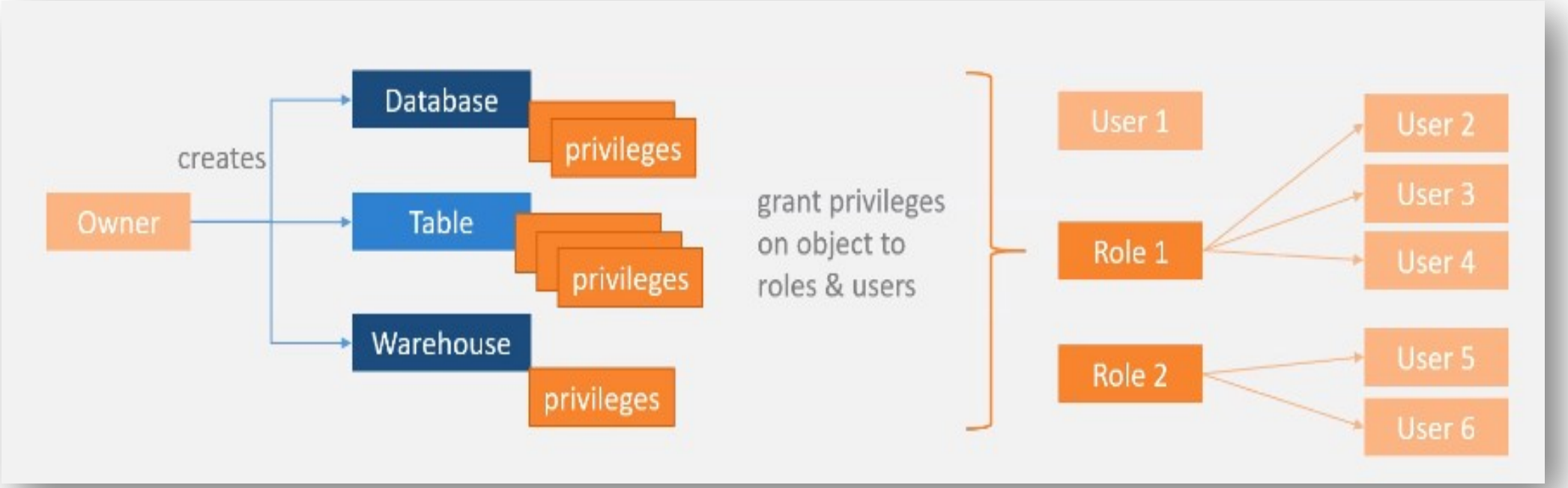
- What is Access Control?
  - Access control determines who can access database objects and perform operations on specific objects in Snowflake.
- Snowflake supports and combines both of below access control models.
  - **Discretionary Access Control (DAC):** Each object has an owner, who can in turn grant access to that object.
  - **Role-based Access Control (RBAC):** Access privileges are assigned to roles, which are in turn assigned to users.

Key Concepts

Below are the key concepts of Snowflake Access Control mechanism.

- **Securable object:** An entity to which access can be granted. Tables, schemas, views etc.
- **Role:** An entity to which privileges can be granted. Roles are in turn assigned to users. Note that roles can also be assigned to other roles, creating a role hierarchy.
- **Privilege:** It is level of access that can be granted to any object. Like select, create, drop, insert etc.
- **User:** Specifies the people or system to whom the access granted.

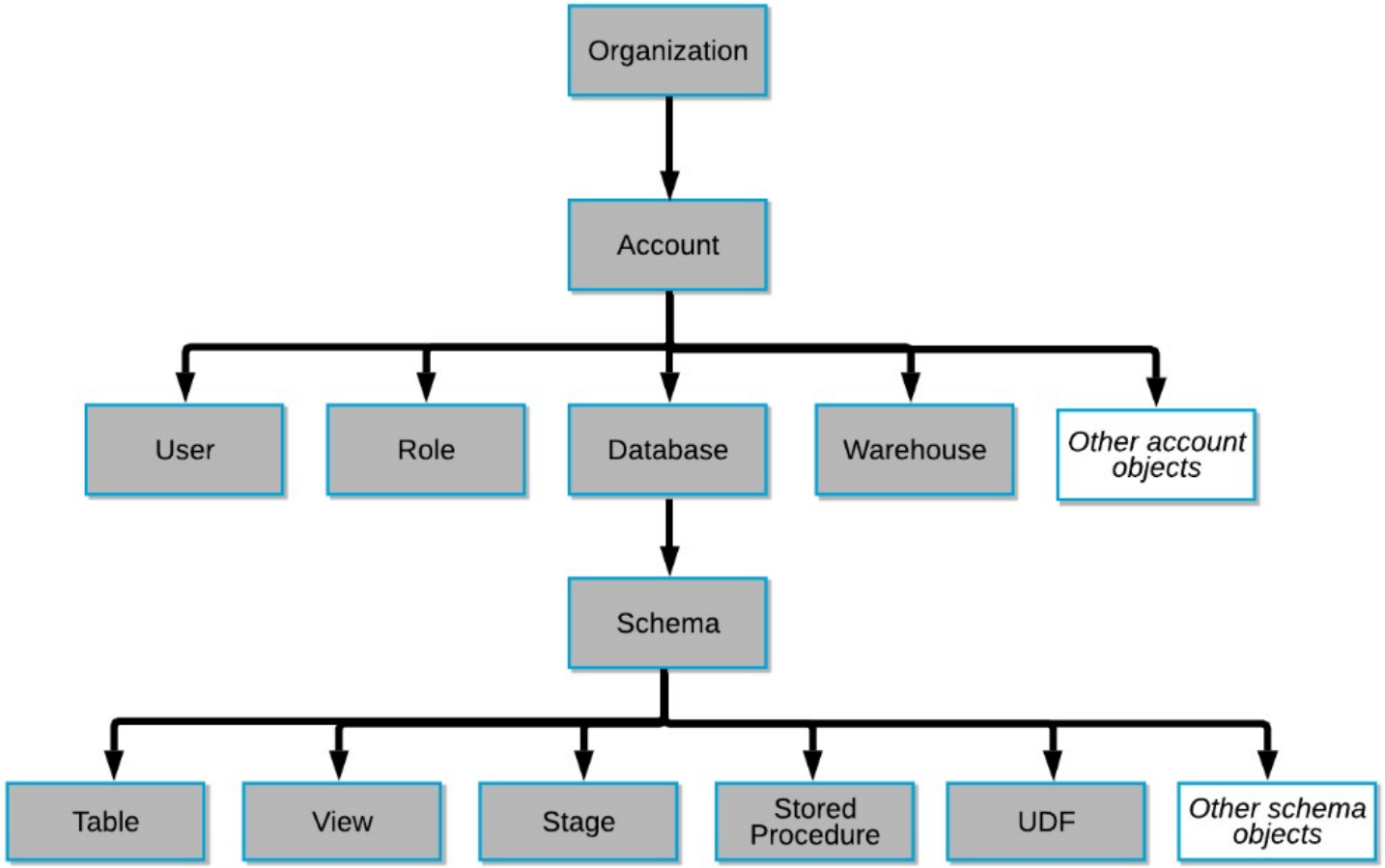
Key Concepts



Privileges

Privilege	Usage
SELECT	Execute a SELECT statement on the table.
INSERT	Execute an INSERT query on the table.
UPDATE	Execute an UPDATE query on the table.
TRUNCATE	Execute a TRUNCATE query on the table.
DELETE	Execute a DELETE query on the table.
ALL [PRIVILEGES]	Grant all privileges, except OWNERSHIP, on the table.
OWNERSHIP	Grant full control over a table.

Object Hierarchy



Roles in Snowflake

- Roles are the entities to which privileges on securable objects can be granted and revoked.
- Roles are assigned to users to allow them to perform actions required for business functions
- A user can be assigned multiple roles. This allows users to switch roles.
- On Snowflake web UI window the top right corner shows which Role is currently we are using.
- Two types of roles
    1. System defined roles
    2. Custom roles

System Defined Roles

1. ORGADMIN (Organization Administrator):

- Role that manages operations at the organization level. More specifically, this role:
- Can create accounts in the organization.
- Can view all accounts in the organization (using SHOW ORGANIZATION ACCOUNTS) as well as all regions enabled for the organization (using SHOW REGIONS).
- Can view usage information across the organization.

2. ACCOUNTADMIN (Account Administrator):

- It is the top-level role in the system.
- Only Account Admin can see all account related things like usage, billing, users, roles, sessions and reader account details.
- Should be granted only to a limited number of users in your account.
- Can enable multi factor authentication.
- Encapsulates the SYSADMIN and SECURITYADMIN system-defined roles.

System Defined Roles

3. SECURITYADMIN (Security Administrator):

- Role that can manage any object grant globally,
- Can create, monitor, and manage users and roles.
- Limited access to Account tab.
- Inherits the privileges of the USERADMIN role via the system role hierarchy.

4. USERADMIN (User and Role Administrator):

- Role that is dedicated to user and role management only.
- This role is granted the CREATE USER and CREATE ROLE security privileges.
- Can create users and roles in the account.

System Defined Roles

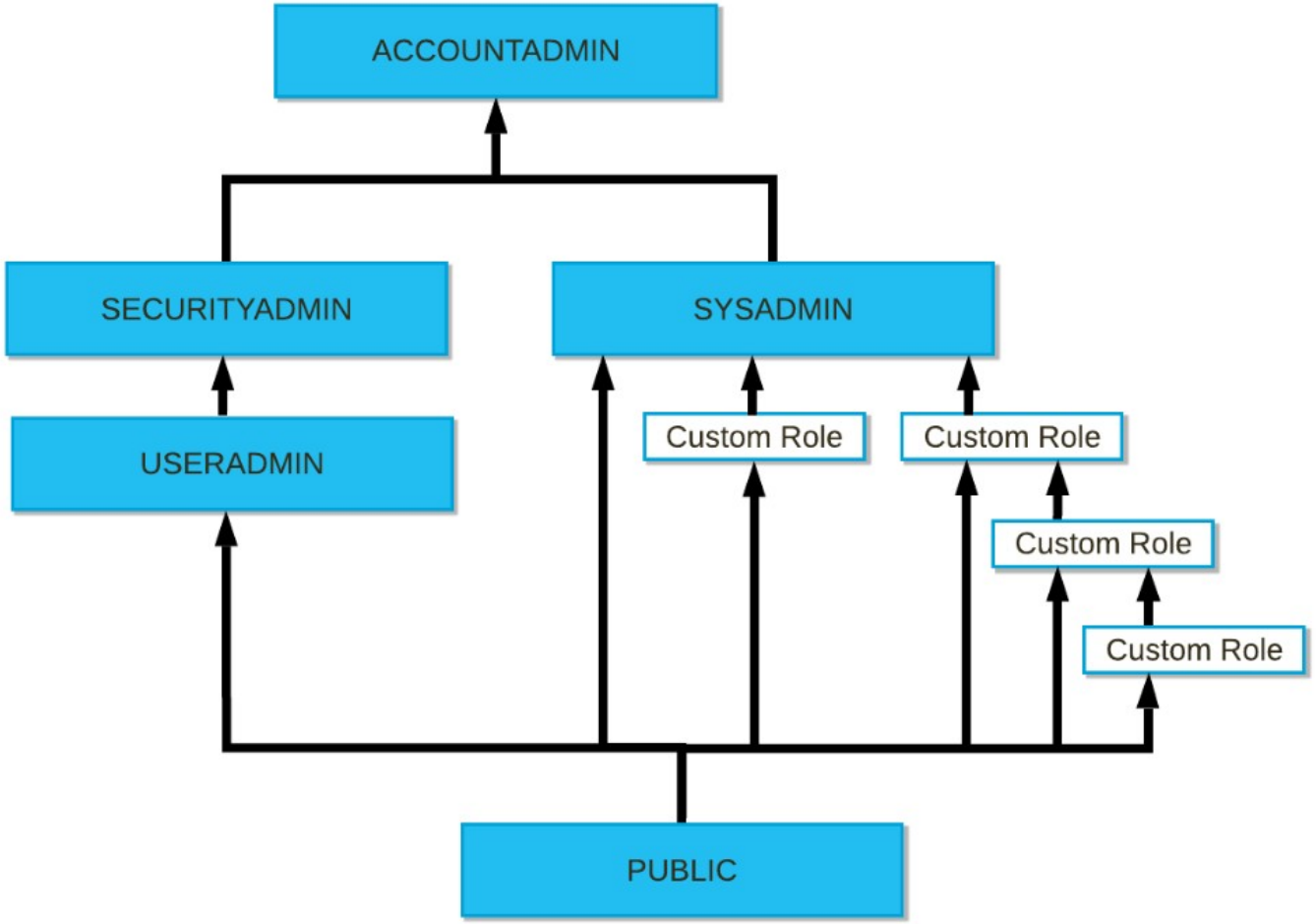
5. SYSADMIN (System Administrator):

- Role that has privileges to create warehouses and databases and other objects in an account.
- This role also has the ability to grant privileges on warehouses, databases, and other objects to other roles.
- All custom roles are assigned to the SYSADMIN role.

6. PUBLIC:

- Automatically granted to every user and every role in your account.
- The objects owned by the role are, by definition, available to every other user and role in the account.
- This role is typically used in cases where all users have all access to the objects.

Role Hierarchy



Custom Roles

- Custom roles can be created by the USERADMIN role (or a higher role) as well as by any role to which the CREATE ROLE privilege has been granted.
- By default, a newly-created role is not assigned to any user, nor granted to any other role.
- Snowflake recommends creating a hierarchy of custom roles, with the top-most custom role assigned to the system role SYSADMIN. This will allow system administrators to manage all objects in the account.
- If custom role is not assigned to SYSADMIN through a role hierarchy, the system administrators will not be able to manage the objects owned by that role. Only SECURITYADMIN can view the objects and modify their access grants.

Thank You